Toronto Metropolitan University

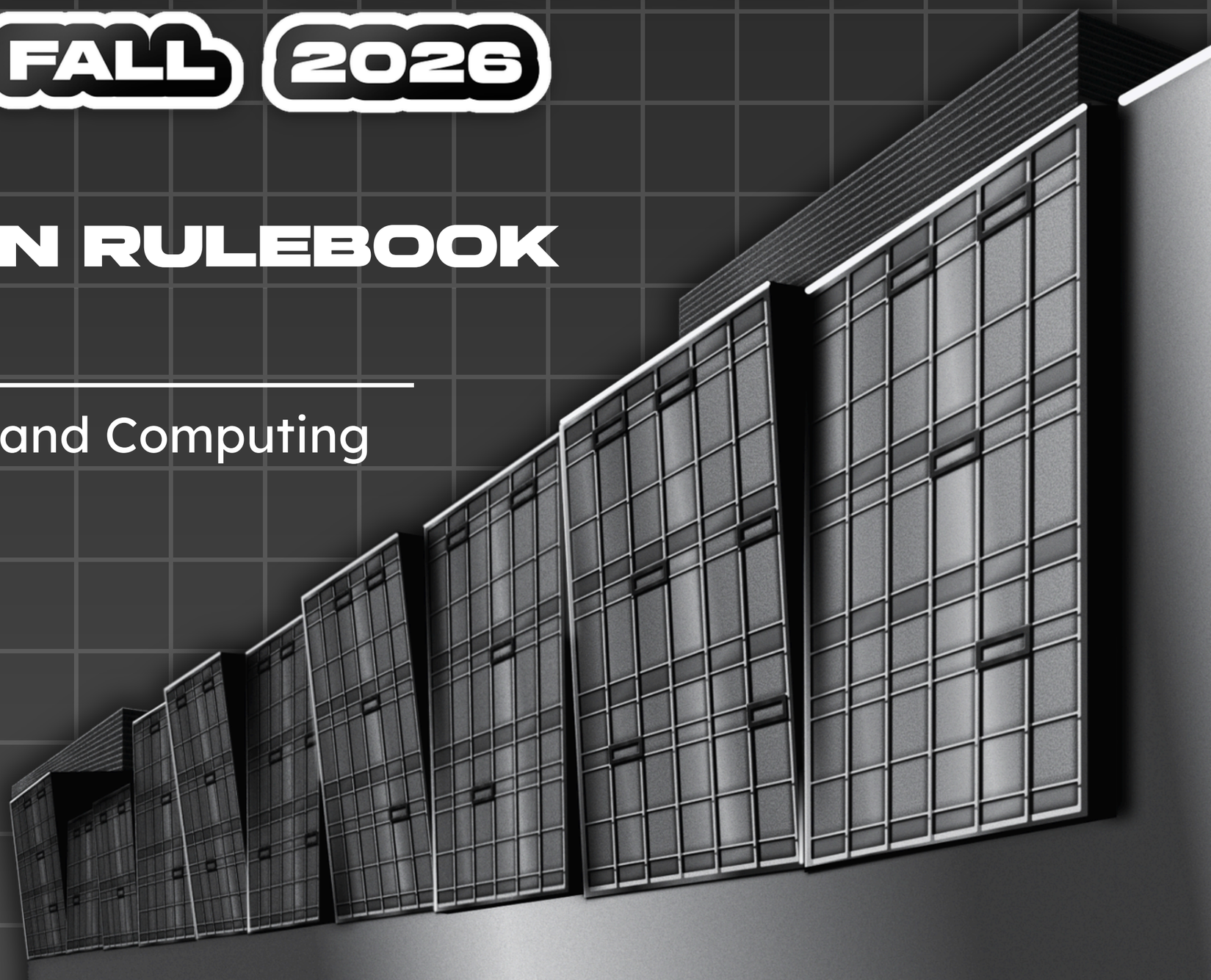
Metropolitan Undergraduate Engineering Society

TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION FALL 2026

SENIOR DESIGN RULEBOOK

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre



1. OVERVIEW

The Senior Design Competition challenges engineering students to apply advanced engineering principles, creativity, and problem-solving skills to develop a functional prototype addressing a practical engineering problem.

Competitors must integrate technical knowledge, engineering design, and effective teamwork while operating under time, material, and resource constraints.

Teams will be evaluated on the quality of their engineering decisions, the functionality and reliability of their prototypes, their ability to satisfy competition requirements, and their communication and technical reasoning.

The specific engineering challenge, design requirements, and testing procedures will be communicated through the official Senior Design Competition Abstract.

1.1. TEAM COMPOSITION

Each Senior Design team must consist of a minimum of three (3) and a maximum of four (4) competitors.

Competitors must satisfy the eligibility requirements established for the Senior Design category under the official MEC competition regulations.

1.2. LANGUAGE

Presentations and supporting presentation materials may be prepared in either English or French. Teams must use their selected language consistently throughout their presentation and associated materials.

1.3. RESOURCES

1.3.1. Facilities Required

The following facilities will be provided:

- One (1) designated presentation area for the competition briefing and team presentations.
- One (1) large workroom divided into team workspaces, or separate workrooms for participating teams.
- One (1) designated prototype testing area.
- Appropriate electrical access for approved equipment.

1.3.2. Personnel Required

1.3.2.1. Judges

A minimum of three (3) judges, with an odd number of judges overall, will assess the engineering quality, prototype functionality, technical decisions, and presentation performance of competing teams.

Judges should possess appropriate engineering knowledge or relevant technical experience.

1.3.2.2. Competition Leads

The Competition Leads, working with the VP Competitions, are responsible for administering the competition, presenting the problem statement, communicating official clarifications, coordinating testing, and answering authorized questions.

Only the Competition Leads may provide official interpretations of competition requirements.

1.3.2.3. Official Timekeeper

The Official Timekeeper is responsible for monitoring and enforcing the time limits established for the development period, presentations, and question periods.

1.3.3. Equipment Supplied by MEC

During the design phase:

- One (1) designated work area per team.
- At least one (1) table and four (4) chairs.
- Paper and writing materials.
- Approved construction tools and equipment.
- Access to electrical power where required.
- Access to the MEC Shop for approved materials and components.

During the presentation:

- One (1) presentation area.
- One (1) projector or equivalent display.
- One (1) table for displaying the completed prototype.
- Appropriate presentation equipment.

1.3.4. Allowed External Resources

Competitors may bring textbooks, course notes, technical references, and other permitted reference materials.

Each competitor may use one (1) personal electronic device at a time.

Internet access may be used for permitted research, technical documentation, and programming assistance.

External construction materials, preassembled mechanisms, and unauthorized equipment may not be incorporated into the competition prototype.

Presentation templates may be prepared before the competition, provided they contain no completed challenge-specific technical content.

1.4. PROCEDURE / TIMELINE

1.4.1. Pre-Competition

Prior to the competition, official rules, schedules, and relevant information will be communicated through authorized MEC communication channels.

1.4.2. Competition

1.4.2.1. Presentation of Problem

The official engineering problem will be presented to all competitors at the beginning of the competition.

The Competition Leads will explain the objectives, requirements, restrictions, available resources, and evaluation procedures.

1.4.2.2. Question Period

Competitors will have fifteen (15) minutes to ask questions following the initial presentation of the problem.

Official clarifications will be communicated consistently to all teams.

1.4.2.3. Development of the Solution

Teams will have ten (10) hours to develop their solutions, construct their prototypes, perform permitted testing, and prepare the required deliverables.

The Competition Organizers will announce the official start time, submission deadline, and testing schedule.

All required deliverables must be submitted before the announced deadline.

1.4.2.4. Rest Period

Rest periods and breaks will be scheduled by the Competition Organizers and communicated to all teams. Competitors must follow the official competition schedule.

1.4.2.5. Presentation and Tests

Each team will be allocated twenty-five (25) minutes for evaluation, consisting of ten (10) minutes for the engineering presentation, ten (10) minutes for official prototype testing, and five (5)

minutes for questions from the judging panel. All teams must follow this sequence and complete their evaluation within the allocated time.

All team members must be present and participate.

The completed prototype must be available for inspection and functional evaluation.

Official prototype testing will take place during the ten-minute testing portion of each team's allocated evaluation period.

All teams will be evaluated under equivalent testing procedures.

1.4.3. Timekeeping

1.4.3.1. During the Design Phase

The development period begins when all teams have reached their assigned work areas and the Competition Leads authorize the start.

Remaining time will be announced two (2) hours, one (1) hour, thirty (30) minutes, and ten (10) minutes before the end of the development period.

1.4.3.2. During the Presentation

Competitors will be notified when two (2) minutes and one (1) minute remain.

A visual countdown will be provided during the final thirty (30) seconds.

Presentations are limited to ten (10) minutes. A fifteen (15) second grace period may be provided solely to conclude the final statement. Teams must stop presenting when instructed by the Official Timekeeper.

Following the ten-minute presentation, teams will proceed to the ten-minute prototype testing period. The judging panel will then have five (5) minutes for questions.

1.4.4. Presentation Order

- Presentation order will be determined randomly.
- The order will be announced at least thirty (30) minutes before presentations begin.
- Teams may not exchange presentation positions without authorization.
- Competitors may not observe another team's presentation before completing their own.
- Prototype testing may follow a separate schedule established by the Competition Organizers.

1.4.5. Deliverables

Each team must develop a functional physical prototype and prepare the required technical presentation.

Teams must also submit any additional electronic files or supporting documentation specified in the official Competition Abstract.

All required deliverables must be submitted before the announced deadlines.

The completed prototype must be available for official inspection and testing.

1.4.6. Response to Questions

Only the Competition Leads may provide official clarification of competition rules, requirements, and procedures.

Questions submitted during the development period must be directed through the authorized clarification process.

Clarifications that materially affect the competition will be communicated to all teams.

Judges and volunteers must not provide unauthorized technical assistance or interpretations that could give a team an unfair advantage.

1.5. ASSESSMENT AND JUDGING

- The judging panel should consist of an odd number of judges whenever practical.
- A minimum of three (3) judges is required.
- Teams will be evaluated using the official Senior Design judging matrix.
- All teams will be assessed using consistent evaluation criteria.
- Feedback will be provided according to the official MEC competition procedures.

1.6. SOLUTION AND RUBRIC

The winning solution will not necessarily be the most technically complex design. Judges will consider engineering quality, functionality, practicality, reliability, and the ability to satisfy the competition requirements.

A simpler, reliable solution may receive a higher score than a more complex design that does not function consistently.

1.6.1. Judging Matrix

Category	Judging Criteria	Points
Design Quality	Engineering design, structural quality, creativity, and technical integration	/10
	Functionality-focused mechanical design and practical engineering solutions	/10
	Quality of technical implementation and safe construction	/5
	Engineering decisions, resource efficiency, and design trade-offs	/5
Design Quality Subtotal		/30
Functional Testing	Performance against the primary competition objectives	/15
	Effectiveness of user interaction or system operation	/5
	Implementation of required supporting functionality	/5
	Performance of the required system integration and reliability tests	/10
	Quality of operational feedback and system response	/5
Functional Testing Subtotal		/40
Presentation	Engineering explanation and design justification	/15
	Presentation quality and visual aids	/10
	Technical understanding and responses to questions	/5

Presentation Subtotal		/30
TOTAL		/100

2. VIOLATION OF COMPETITION RULES

2.1. REPORTING AND APPEALS PROCEDURE

In the event of a suspected rule violation, the following process shall occur:

i.

The **Rule Violation Reporting System** shall be:

- A form created by the **Rule Violation Committee (RVC)**.
- Made accessible to all competitors during MEC and for at least **24 hours afterwards**.

ii.

The **Rule Violation Form** shall include the following fields:

- **a.** Name of Reporting Individual
- **b.** Competition/Position of Reporting Individual
- **c.** Alias of Reporting Individual (if applicable)
- **d.** Rule Violation Date and Time
- **e.** Name of the Individual Whom the Rule Violation Report is Against
- **f.** Rule Violation Details (including references to rulebook sections and backup information, if applicable)
- **g.** Rule Violation Witness(es), if applicable
- **h.** Contact Information

iii.

The competitor(s) shall be notified:

- **Verbally** and **in writing**
- **Immediately** upon receipt of a rule violation submission against them
- They shall be allowed to **continue the competition and work on their solution**

iv.

Competitor(s) shall be provided, **in writing**, the alleged violation of the rules following the **design phase**.

v.

If the competitor(s) are **found guilty** of a violation following the conclusion of the design phase:

- They will have **one (1) hour** to appeal the RVC's decision to the **Rule Violation Appeal Committee (RVAC)**.

a.

Appeals must be submitted:

- Within **one (1) hour** of receiving the RVC's decision,
- Or by **9:00 AM the following morning** if the decision was made after **12:00 AM**.

vi.

In the event of a rule violation discovery **after the competition**:

- Reports should be made **via email to the Chair(s)**, following the same form requirements listed above.
- Competitor(s) will be **notified upon receipt** of the rule violation submission.
- The **RVC** shall provide a report with their findings to the competitor(s) within **48 hours**.
- Appeals to post-competition submissions must be made to the **RVAC within 48 hours**.